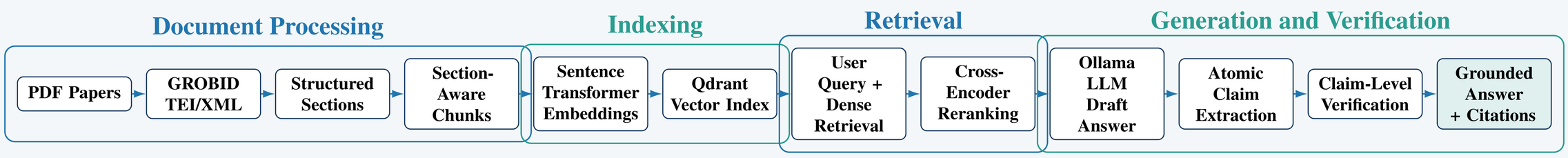
Research Paper Assistant using Retrieval-Augmented Generation (RAG)

A Multi-Paper Scientific QA System with Claim-Level Hallucination Verification

LOCAL EVIDENCE-FIRST RAG

End-to-end architecture



Problem and objective

Researchers must locate evidence across long, structured papers. General-purpose language models can answer from outside knowledge or introduce unsupported claims.

Objective: answer single-paper and cross-paper questions using only retrieved evidence, with citations and claim-level filtering before synthesis.

- Ingest multiple scientific PDFs.
- Preserve document structure and provenance.
- Retrieve and rerank relevant passages.
- Compare evidence across named papers.
- Remove claims not supported by retrieved excerpts.

Scientific PDF ingestion

GROBID converts PDFs into structured TEI/XML. The parser retains title, authors, abstract, sections, subsections, and paragraphs.

Structure-aware extraction gives downstream chunks meaningful section context instead of treating a paper as an undifferentiated text stream.

Section-aware semantic indexing

Each chunk retains:

- paper title and document name,
- section name and chunk ID,
- scientific text.

Embedding model	all-MiniLM-L6-v2
Vector dimension	384
Vector database	Qdrant
Similarity	Cosine similarity

Dense vectors support semantic search while metadata preserves where every passage originated.

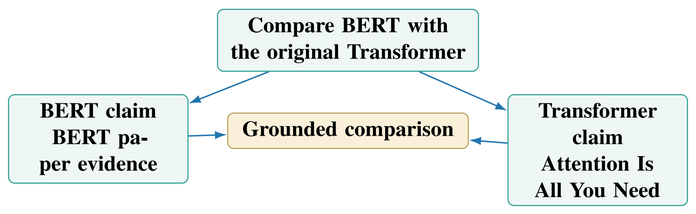
Retrieval and reranking



Dense retrieval supplies candidates. The `ms-marco-MiniLM-L-6-v2` cross-encoder then scores each question-passage pair and improves evidence ordering.

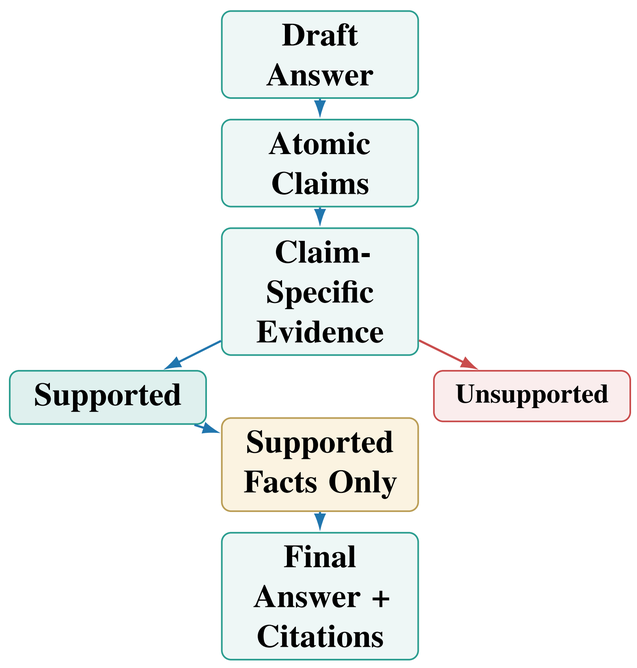
Multi-paper comparison

Naive global search can over-represent one paper or retrieve a passage that only mentions the target model. The system resolves the primary subject and routes atomic claims to the relevant paper before claim-specific retrieval.



Routing signals: atomic decomposition, entity detection, primary-subject identification, referenced-entity distinction, paper filters, and claim-specific retrieval.

Claim-level hallucination reduction



The final synthesizer receives verified facts rather than the draft. Source IDs, paper scope, entity compatibility, duplicates, and malformed partial verdicts are validated before a fact can reach the answer.

Verification example

Rejected draft claim

"The original Transformer uses constrained self-attention."

The evidence describes masking in the decoder; it does not establish that the entire architecture uses constrained self-attention. **Removed.**

Supported claim

"BERT uses bidirectional self-attention."

Explicitly supported by the BERT model-architecture evidence. **Retained.**

Grounded comparison example

Question: Compare BERT with the original Transformer architecture.

BERT

- Multi-layer bidirectional Transformer encoder.
- Uses bidirectional self-attention.
- Unified architecture across tasks.

Original Transformer

- Attention-based sequence-transduction model.
- Encoder-decoder architecture.

Key difference: BERT is encoder-based; the original Transformer has both encoder and decoder components.

[Source: BERT, Model Architecture] [Source: Transformer, Model Architecture]

Evaluation framework implemented

No completed evaluation run was available in `evaluation/results/`; therefore, no performance numbers or charts are shown.

Retrieval evaluation

- Hit@1, Hit@3, Hit@5, and mean reciprocal rank.
- Vector retrieval before vs. after reranking.
- Paper and section matching.
- Comparison-paper coverage.

Grounding evaluation

- Fully, partially, and unsupported claim rates.
- Citation validity and rejected-claim leakage.
- Expected-fact coverage and abstention accuracy.

18 Evaluation questions	**3** Cross-paper comparisons
2 Abstention tests	**0** Fabricated metrics

Implemented contributions

- Structure-aware scientific PDF ingestion.
- Section-aware semantic indexing with provenance.
- Dense retrieval plus cross-encoder reranking.
- Paper-aware multi-paper evidence routing.
- Atomic claim extraction and supported-core recovery.
- Claim-level semantic and source-scope verification.
- Deduplicated, citation-grounded answer synthesis.
- Reusable retrieval and generation evaluation framework.

Technology stack

Language	Python
Parsing	GROBID, TEI/XML, PyMuPDF
Embeddings	Sentence Transformers
Vector store	Qdrant
Reranking	Cross-Encoder
Local LLM	Ollama, Llama 3.2 3B
Workflow	Docker, Jupyter Notebook

Limitations and future work

Current limitations

- Local LLM verification can make semantic judgment errors.
- Dense retrieval can miss highly specific technical evidence.
- Claim verification increases inference time.
- The initial evaluation set is intentionally small.

Next steps

- Hybrid dense + BM25 retrieval.
- NLI-based and batched claim verification.
- Larger evaluation set and paper collection.
- Streamlit user interface.

Conclusion

The system combines structured scientific document processing, semantic retrieval, reranking, and claim-level verification to produce evidence-grounded answers across multiple research papers.